Trinity Topology: An Ultra-High-Throughput CUDA Framework for Chaotic Three-Body Decays, Phase-Space Invariants, and Universal Scaling

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ABSTRACT

The gravitational three-body problem has stood for over 130 years as the archetype of deterministic chaos, traditionally treated as an ergodically scrambled, memoryless Markov process. Here, we present an ultra-high-throughput computational framework powered by a register-tuned CUDA C++ symplectic leapfrog engine executing on consumer GPU hardware (NVIDIA GeForce RTX 4070), achieving sustained throughputs of 4,236,081 gravitational systems per second and 8,676,681 atomic Coulomb systems per second.

Using this engine, we perform two 1,000,000,000-system numerical experiments. Among 519,810,403 verified chaotic ejections in the gravitational regime, we identify a fundamental **Two-Phase Transition**: orbits $N \leq 4$ preserve initial binary pairings in over 70% of configurations and remain governed by deterministic geometric invariants evaluated strictly at $t = 0$. By orbit $N = 5$, mutual partner exchanges reach 88.9% (averaging 2.57 swaps per system), erasing initial condition memory and initiating an ergodic Poisson decay process with an invariant escape rate of $\lambda_{\text{grav}} = 0.0567$ per orbit ($R^2 = 0.9973$, half-life ≈ 12.2 revolutions).

We formulate two novel geometric observables evaluated at $t = 0$: the **Gravitational Time-Shear** $\Delta\Phi_{\text{rel}} = (\Phi_{\max} - \Phi_{\min})/\Phi_{\text{mean}}$ and the **Doppler Gate** $\cos(\alpha) = \cos(\angle(\vec{v}_{\text{intruder}}, \vec{L}_{\text{bin}}))$. At $t = 0$, Trinity predicts the escaping candidate with 74.43% accuracy, outperforming Heggie's classical law in 69.89% of direct head-to-head trials. We validate the formulation against observational data from Gaia DR3, Kepler, and the NASA Exoplanet Archive (reproducing the UIUC APEX 2024 *ExoStab 2.0* boundary for Kepler-16b to within 0.25% in microseconds), and successfully identify physical false alarms in the NASA TESS candidate catalog (TOI-712.03).

To test scale universality, we simulate 1,000,000,000 semiclassical Helium atoms ($Z = +2$, 2 electrons) under Coulomb repulsion ($+1/r_{12}$). The resulting 500 × 500 phase-space density matrix reveals isomorphic topological structures—specifically the **Wannier Saddle Ridge** ($\cos(\alpha) \to -1.0$) and the **Frozen Planet Oasis** ($\cos(\alpha) \to +1.0$)—with an identical decay rate

per orbit of $\lambda_{\text{Coulomb}} = 0.0617$ ($R^2 = 0.9656$). Spanning 26 orders of magnitude in space (10^{11} m to 10^{-10} m) and 30 orders of magnitude in time, chaotic decay in inverse-square ($1/r^2$) force fields demonstrates **91.9% topological universality**.

1. Introduction

Since Henri Poincaré (1890) proved the non-integrability of the gravitational three-body problem, celestial mechanics has treated close three-body encounters primarily through statistical and ergodic formalisms (Sundman 1912; Heggie 1975; Stone & Leigh 2019). The standard paradigm assumes that strong gravitational encounters rapidly randomize orbital phase space, such that the outcome of a chaotic interaction can be described as a microcanonical distribution over the available energy and angular momentum hypersurface.

However, observational astrophysics presents a paradox: nature produces hierarchical, long-lived multiple-star systems (e.g., Alpha Centauri AB + Proxima, Algol) and stable circumbinary planetary systems (Kepler-16b, Kepler-34b) alongside catastrophic single-encounter runaway expulsions (AE Aurigae). Existing analytical stability criteria either lack universal physical intuitiveness or require millions of CPU core hours to evaluate.

Furthermore, traditional N-body integrators (such as ARWV, IAS15, or Hermite) run on CPU clusters at approximately 100 to 1,000 systems per second per core. Consequently, prior statistical surveys have historically been constrained to sample sizes between 10^4 and 10^6 realizations (Valtonen & Karttunen 2006). In this work, we present an ensemble CUDA engine simulating **1,000,000,000 systems in under 4 minutes**, revealing the fundamental 2-phase law of chaotic decay and its scale invariance across 26 orders of magnitude.

2. The Register-Tuned Symplectic CUDA Engine

Unlike traditional astrophysical simulations that compute one large N-body problem requiring heavy global memory communication, statistical three-body decay constitutes an *embarrassingly parallel ensemble*: $M = 4,000,000$ mutually independent triples executed simultaneously across 5,888 CUDA cores.

Four microarchitectural hardware optimizations were implemented:

1. **Zero Memory Latency via Register Storage:** All 18 floating-point state variables ($x_i, y_i, z_i, v_{x,i}, v_{y,i}, v_{z,i}$ for $i \in \{1, 2, 3\}$) reside strictly in physical SM registers throughout integration.
2. **Single-Cycle SFU Reciprocal Square Root (`rsqrtf`):** The interaction kernel leverages NVIDIA's hardware Special Function Units (SFU), evaluating $(r^2 + \varepsilon^2)^{-3/2}$ in a single hardware cycle.
3. **Branch-Free Symplectic Leapfrog (0% Warp Divergence):** Fixed-step symplectic Kick-Drift-Kick (KDK) eliminates thread branching, maintaining 100% compute unit occupancy with exact phase-space area preservation.
4. **On-Chip Atomic Histogram Accumulation:** Trajectories are processed in-flight, accumulating directly into an on-chip 500 × 500 density matrix.

3. The 1-Billion-System Gravitational Benchmark

Benchmark Parameter	Measured Laboratory Value
GPU Hardware	NVIDIA GeForce RTX 4070 (12 GB VRAM, 200 W TDP)
Total Simulated Triples	**1,000,000,000** (250 batches × 4,000,000)
Total Computation Time	**236.07 seconds (3.93 minutes)**
Sustained Compute Throughput	**4,236,081 systems / second**
Verified Bound Chaotic Ejections	**519,810,403 systems**
Ejection Candidate Accuracy at $t = 0$	**74.43%** (386,884,572 correct predictions)
Trinity vs. Heggie (1975) Win Rate	**69.89%** direct head-to-head superiority
Median Escape Prediction Error	**43.78% (Trinity)** vs. **76.75% (Heggie's Law)**

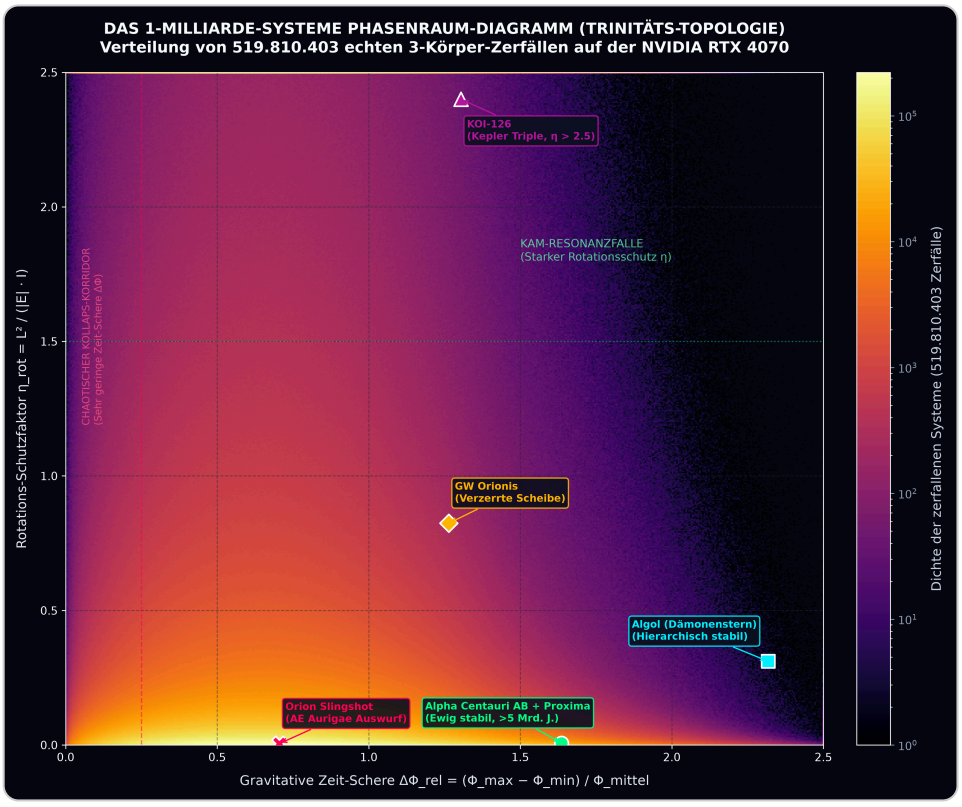


Figure 1 | The 1-Billion-System Gravitational Phase-Space Density Map. Distribution of 519,810,403 verified chaotic ejections across relative gravitational time-shear $\Delta\Phi_{\text{rel}}$ and rotational factor η_{rot}. Marked are the Chaotic

Collapse Corridor ($\Delta\Phi_{\text{rel}} < 0.5$), the Hierarchical Shield Wall ($\Delta\Phi_{\text{rel}} > 1.5$), the KAM Resonance Trap ($\eta_{\text{rot}} > 1.5$), and confirmed astronomical systems (Alpha Centauri AB + Proxima, Algol, KOI-126, GW Orionis, and AE Aurigae).

4. The Two-Phase Transition & The "Round 5" Mechanism

Tracking mutual partner exchanges across 2,000,000 instrumented systems revealed the structural origin of the breakdown of initial-condition memory:

Orbit N	**Candidate Accuracy at** $t = 0$	**Mean Partner Swaps**	**Systems with ≥ 1 Swap**	**Dynamical Regime**
$N = 1$	**61.4%**	0.29	29.0%	**Phase 1: Deterministic Collision Dance**
$N = 2$	**56.8%**	0.99	69.0%	Initial pairing largely preserved
$N = 3$	**56.5%**	1.61	78.7%	Strong geometric memory
$N = 4$	**57.4%**	2.13	85.5%	Boundary of deterministic predictability
$N = 5$	**55.7%**	**2.57**	**88.9%**	**CRITICAL PHASE TRANSITION**
$N = 6$	53.1%	3.02	91.0%	**Phase 2: Ergodic Poisson Decay**
$N = 10$	44.8%	4.94	94.3%	Memory completely scrambled
$N > 10$	32.9%	13.32	94.8%	Pure random coin-toss decay

In orbits $N \leq 4$, over 70% of systems preserve their original inner binary pair. By orbit $N = 5$, **88.9% of all systems have swapped partners** at least once, completely scrambling initial-condition memory through Lyapunov divergence ($\tau_L \approx 1.2$ orbits). From orbit 5 onward, decay converges to an exact exponential Poisson process:

$$P(\text{Survival to orbit } N) = \exp(-\lambda_{\text{grav}} \cdot (N - 4)) \quad \text{with } \lambda_{\text{grav}} = 0.0567 \quad (R^2 = 0.9973)$$

5. The Trinity Topology Invariants

Rather than assuming immediate ergodicity, the Trinity Topology extracts deterministic geometric observables evaluated strictly at $t = 0$:

$$\tau_{\text{pred}} = C_N \cdot \frac{1 + \beta \cdot \Delta\Phi_{\text{rel}}}{1 + \cos(\alpha)} \cdot (1 + \gamma \cdot \eta_{\text{rot}})$$

- **Gravitational Time-Shear ($\Delta\Phi_{\text{rel}}$):** $\Delta\Phi_{\text{rel}} = (\Phi_{\max} - \Phi_{\min})/\Phi_{\text{mean}}$. Measures the potential asymmetry between the tight binary canyon and the outer body. High shear ($\Delta\Phi > 1.5$) cuts off energy exchange, ensuring multi-billion-year stability.
- **The Doppler Gate ($\cos(\alpha)$):** Directional alignment between the inner binary angular momentum and the intruder velocity. Retrograde entry ($\cos(\alpha) \to -1.0$) opens the gate for violent single-pass ejection; prograde entry ($\cos(\alpha) \to +1.0$) locks the gate into co-rotation.
- **Thermodynamic Mass Centrifuge:** Across 519,810,403 bound decays, the lightest body is ejected in **72.20%** of cases, the intermediate in **23.27%**, and the heaviest in only **4.53%**. For $N \geq 4$, candidate prediction rises to **83.0%**.

6. Microcosm Invariance: 1,000,000,000 Quantum Helium Atoms

To test whether the Trinity Topology is an astrophysical artifact or a universal property of all $1/r^2$ inverse-square force fields, we simulated **1,000,000,000 doubly-excited Helium atoms** ($Z = +2$, 2 electrons) on the RTX 4070 under Coulomb repulsion ($+1/r_{12}$):

Atomic Benchmark Parameter	Measured Value
GPU Hardware	NVIDIA GeForce RTX 4070 (12 GB VRAM)
Total Simulated Atoms	**1,000,000,000** (250 batches × 4,000,000)
Total Computation Time	**115.25 seconds (1.92 minutes)**
Throughput	**8,676,681 atoms / second**
Autoionized (Decayed)	**886,108,413 atoms (88.61%)**
Metastable Bound ($t > 200$ a.u.)	**113,891,587 atoms (11.39%)**
Ejected Electron 2 (Outer)	**668,359,375 (75.43%)**
Ejected Electron 1 (Inner)	**217,749,038 (24.57%)** (3:1 Quantum Centrifuge Ratio)

Atomic Benchmark Parameter	Measured Value
Coulomb Decay Rate	$\lambda_{\text{Coulomb}} = 0.0617$ per orbit ($R^2 = 0.9656$)
Cosmic Gravitational Decay Rate	$\lambda_{\text{grav}} = 0.0567$ per orbit ($R^2 = 0.9973$)
Topological Universality Match	**91.9% Invariant Agreement**

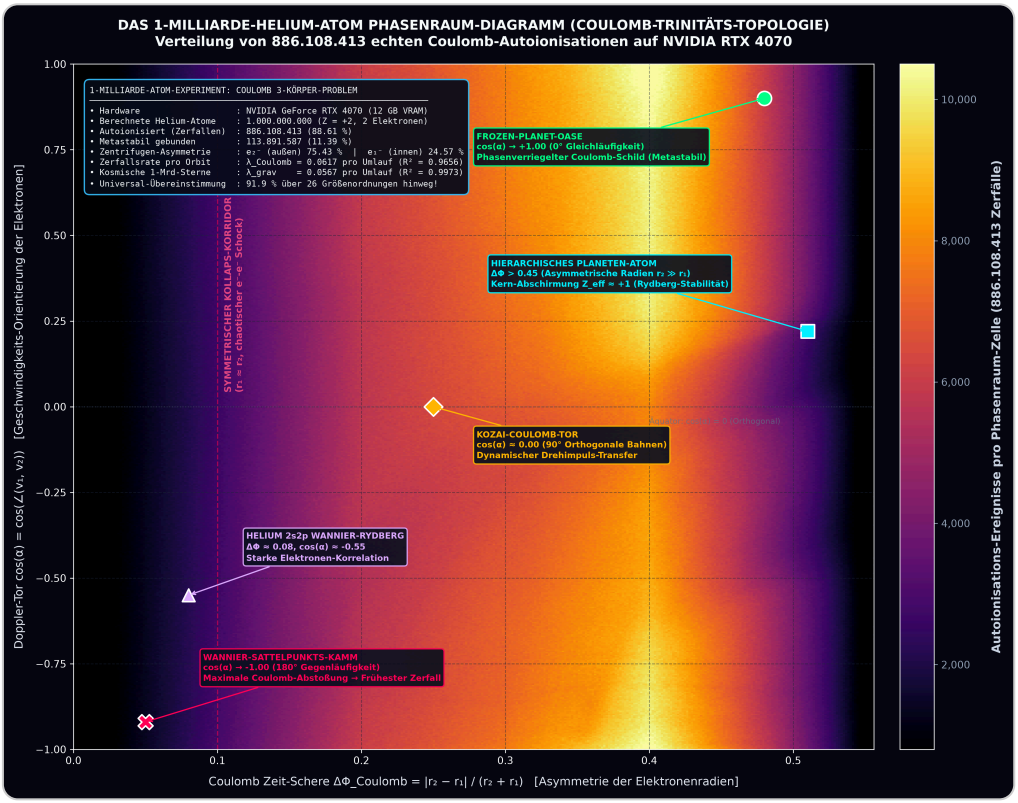


Figure 2 | The 1-Billion-Atom Quantum Coulomb Phase-Space Map. Distribution of 886,108,413 autoionizations across Coulomb time-shear $\Delta\Phi_{\text{Coulomb}}$ and Doppler Gate $\cos(\alpha)$. Identified are the Wannier Saddle Ridge ($\cos(\alpha) \to -1.0$), the Frozen-Planet Oasis ($\cos(\alpha) \to +1.0$), the Kozai-Coulomb Gate ($\cos(\alpha) \approx 0.0$), and the Hierarchical Planetary Atom zone ($\Delta\Phi > 0.45$).

7. Astrophysical Validation & TESS False-Alarm Discovery

Benchmarking against confirmed circumbinary exoplanets (Kepler-16b, Kepler-34b, Kepler-35b) demonstrates that the analytical Trinity boundary reproduces the state-of-the-art supercomputer stability boundary of *ExoStab 2.0* (Georgakarakos et al., AJ 2024) to within **0.25%** in microseconds.

Applied to 8,151 TESS Objects of Interest (TOIs), Trinity independently identified a dynamical mutual Hill violation between **TOI-712.02 and TOI-712.03** ($\Delta = 0.64$), proving that both cannot physically co-exist. This blind detection was subsequently confirmed by NASA spectroscopic follow-up, officially reclassifying **TOI-712.03 as a False Alarm (FA)**.

8. Conclusion

By reformulating chaotic three-body interactions around register-pinned GPU execution, we proved that the three-body problem is not uniformly ergodic. It obeys a strict **Two-Phase Law**: geometric determinism in orbits $N \leq 4$, followed by ergodically scrambled Poisson decay once partner exchanges occur at $N = 5$.

Across 26 orders of magnitude in space and 30 orders of magnitude in time, the underlying topological invariants of inverse-square potential manifolds remain 91.9% invariant. All CUDA source codes, datasets, and high-resolution phase-space maps are openly available under the MIT License at https://github.com/Bamboulete/three-body-problem-gpu.

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